



CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

– Hanoi, Sep 2025–

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I. Record of Changes

Date	A* M, D	In charge	Change Description
16/9/2025	A	All of members	Add new background include: record of change, definition, project introduction
18/9/2025	A	All of members	Add existing systems.
20/9/2025	A	All of members	Add new include: business opportunity, software product vision, project scope & limitations.
12/10/2025	M	All of members	Modify existing system analysis.
13/10/2025	M	Hai	Modify product background.
15/10/2025	M	Quat	Modify project scope & limitations
11/11/2025	M	Hien	Modify major feature diagram
11/12/2025	M	Tuan	Modify existing system analysis, redraw major feature diagram
11/12/2025	M	Binh	Modify Project Scope & Limitations

*A - Added M - Modified D - Deleted

Definitions and Acronyms

Acronym	Meaning
A	Added
M	Modified
D	Deleted
OLP-SkillUp	Online Learning Platform – SkillUp
AI	Artificial Intelligence
MOOC	Massive Open Online Course
WCAG	Web Content Accessibility Guidelines
LMS	Learning Management Systems
HRIS	Human Resource Information Systems
IT	Information Technology
SEO	Search Engine Optimization
UX/UI	User Experience / User Interface
iOS	internet Operating System
PDF	Portable Document Format
SEP490_G44	Software Engineering Project 490 - Group 44

II. Project Introduction

1. Overview

1.1 Project Information

- Project name: Online Learning Platform – SkillUp
- Project code: OLP-SkillUp
- Group name: SEP490_G44
- Software type: Website

1.2 Project Team

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2. Product Background

Students and working professionals in Vietnam increasingly need online learning to improve their skills. However, many international platforms are expensive, use English interfaces, and lack interactive support, making them difficult to access. A survey among peers showed that most learners prefer a localized, user-friendly, and affordable platform. Instructors also face challenges in delivering online courses due to limited supporting tools. From this situation, our team of five final-year students proposes **SkillUp** – an online learning platform that is localized, cost-effective, and enhances connections between instructors and learners.

3. Existing Systems

3.1 Coursera

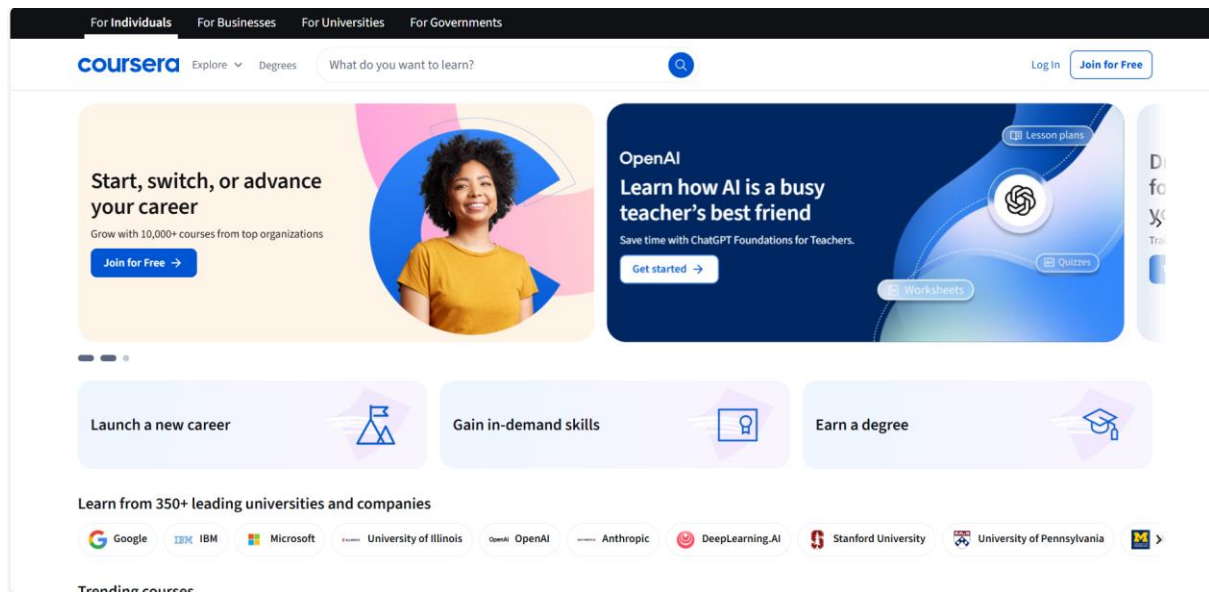


Figure 3.1: Coursera

3.1.1 Description

Coursera, inaugurated in 2012 by Stanford University professors Andrew Ng and Daphne Koller, has established itself as a pioneering force within the massive open online course (MOOC) ecosystem and digital education landscape. The platform operates through strategic collaborative partnerships with over 300 prestigious universities and multinational corporations worldwide, including Stanford University, Yale University, Google, IBM, and Meta. It provides a comprehensively structured hierarchy of educational offerings, encompassing individual skill-development courses, industry-recognized professional certificates, specialized career-focused programs, and fully accredited online degree programs spanning diverse academic disciplines including computer science, business administration, data science, public health, and liberal arts.

Link : <https://www.coursera.org/>

3.1.2 System Actors

The Coursera ecosystem comprises multiple interconnected stakeholders, each fulfilling distinct yet complementary roles within the platform's operational framework:

- **Students/Learners:** Primary end-users who register for educational programs, actively engage with multimedia learning materials, participate in synchronous and asynchronous discussions, submit assessments and capstone projects, complete peer-review assignments, track their academic progress through comprehensive dashboards, and ultimately earn verifiable certificates, specializations, or accredited degrees upon successful completion of prescribed requirements.
- **Instructors/Course Creators:** Academic faculty members, industry practitioners, and subject matter experts responsible for pedagogical design, curriculum development, content creation and curation, learning objective formulation, assessment instrument design, learner engagement facilitation, discussion moderation, and continuous course iteration based on learner feedback and performance analytics.

- **Partner Organizations:** Prestigious universities, research institutions, and corporate entities that contribute rigorously developed educational materials, ensure academic quality assurance, validate certification authenticity, provide institutional credibility, grant academic credits where applicable, and maintain alignment with industry standards and professional competency frameworks.
- **Platform Administrators:** Technical and operational personnel who handle comprehensive platform maintenance, ensure system reliability and security, implement content quality control mechanisms, moderate user-generated content, provide multilingual technical support, resolve learner and instructor queries, manage payment processing, and continuously optimize user experience through data-driven platform enhancements.

3.1.3 Features

Coursera's feature architecture demonstrates sophisticated integration of pedagogical best practices with technological innovation:

- **Multimedia Learning Content:** High-definition video lectures featuring expert instructors, complemented by comprehensive reading materials, scholarly articles, research papers, interactive transcripts with timestamp navigation, and curated external resources that provide contextual depth and facilitate deeper understanding of complex concepts.
- **Diversified Assessment Mechanisms:** Formative assessments including auto-graded quizzes with instant feedback, summative examinations, hands-on programming assignments with automated code evaluation, peer-graded projects fostering collaborative learning, capstone projects requiring real-world application of acquired competencies, and practical labs providing experiential learning opportunities.
- **Credentialing Ecosystem:** Verifiable certificates of completion with blockchain-backed authentication, specialized multi-course programs (Specializations) demonstrating domain expertise, Professional Certificates aligned with industry role requirements, MasterTrack® Certificate programs offering university-level graduate courses, and fully accredited Bachelor's and Master's degree programs conferring official academic credentials.
- **Collaborative Learning Infrastructure:** Threaded discussion forums enabling peer-to-peer knowledge exchange, mentor-moderated study groups, collaborative project workspaces, global learner communities organized by interests and specializations, and integration with professional networking platforms like LinkedIn for credential showcasing.
- **Cross-Platform Accessibility:** Native mobile applications for iOS and Android featuring offline video download capabilities, progress synchronization across devices, push notifications for deadlines and announcements, responsive web interface optimized for various screen sizes, and accessibility features complying with WCAG standards to accommodate diverse learner needs.
- **Adaptive Learning Technologies:** Personalized course recommendations powered by machine learning algorithms analyzing learner behavior, customized learning paths based on prior knowledge and career objectives, and dynamic content sequencing adjusting to individual learning velocities.

3.1.4 Pros

Coursera's competitive advantages manifest across multiple strategic dimensions:

- **Institutional Credibility and Academic Prestige:** Strategic alliances with globally recognized universities (Stanford, Princeton, Yale, Imperial College London) and leading technology corporations (Google, IBM, Meta) confer substantial legitimacy and academic rigor to

educational offerings, significantly enhancing credential market value and employer recognition compared to platforms lacking comparable institutional backing.

- **Comprehensive Content Quality Assurance:** Rigorous content vetting processes, pedagogical design review by educational technology experts, alignment with established academic standards, and continuous quality monitoring mechanisms ensure consistently high-caliber educational experiences that maintain intellectual depth while remaining accessible to diverse learner populations.
- **Extensive Interdisciplinary Coverage:** Remarkably broad curricular scope spanning technology and engineering, business and management, arts and humanities, health and medicine, natural sciences, social sciences, language learning, and personal development, enabling learners to pursue multidisciplinary educational trajectories or pivot across domains to respond to evolving career landscapes.
- **Professionally Valuable Credentials:** Certificates, specializations, and degrees carry substantial weight in professional employment markets due to institutional endorsement, industry alignment, and verifiable skill demonstrations, with many employers specifically recognizing Coursera credentials in hiring decisions, promotional considerations, and professional development requirements.
- **Structured Learning Pathways:** Carefully designed course sequences, prerequisite frameworks, progressive skill development architectures, and guided specialization tracks provide clear roadmaps for learners, reducing decision paralysis and ensuring systematic competency building aligned with career advancement objectives.
- **Financial Aid and Accessibility Initiatives:** Comprehensive financial assistance programs including need-based scholarships, government workforce development partnerships, corporate sponsorship opportunities, and institutional grants democratize access to premium educational content for economically disadvantaged populations globally.

3.1.5 Cons

Despite significant strengths, Coursera exhibits strategic vulnerabilities that constrain its competitive positioning:

- **Premium Pricing Structure:** Subscription-based monetization models (Coursera Plus at \$399-\$599 annually), individual course fees (\$49-\$99), specialization costs (\$39-\$79 monthly), and degree program tuition (\$15,000-\$45,000) create substantial accessibility barriers for price-sensitive market segments, particularly learners in developing economies where educational purchasing power remains limited compared to platforms offering lower-cost or freemium alternatives.
- **Constrained Instructor Interaction:** The massive scale inherent to MOOC architecture fundamentally limits opportunities for personalized instructor engagement, direct mentorship, individualized feedback on complex assignments, and real-time clarification of conceptual difficulties, potentially disadvantaging learners who thrive through interpersonal academic relationships and require customized pedagogical support beyond standardized automated feedback mechanisms.
- **Completion Rate Challenges:** Industry research indicates MOOC completion rates typically range between 5-15%, significantly lower than traditional educational formats, suggesting potential deficiencies in learner motivation sustainment, engagement optimization, accountability mechanisms, or pedagogical effectiveness for certain learner demographics, particularly those lacking intrinsic motivation or structured learning habits.

- **Standardized Pedagogical Approach:** Academic orientation toward theoretical foundations and comprehensive coverage may limit immediate practical applicability for learners seeking rapid skill acquisition for specific job functions, potentially creating misalignment between learning experiences and workplace performance requirements in fast-evolving technical domains.
- **Geographic Content Bias:** Despite international partnerships, course content predominantly reflects Western academic perspectives, pedagogical approaches, and cultural contexts, potentially limiting relevance and engagement for learners from non-Western educational traditions or those seeking localized, culturally contextualized learning experiences.

3.1.6 AI Implementation:

Coursera has strategically integrated artificial intelligence technologies across multiple platform dimensions to enhance scalability, personalization, and pedagogical effectiveness:

- **Automated Assessment Evaluation:** Sophisticated natural language processing algorithms evaluate text-based assignments, essays, and written responses according to instructor-defined rubric criteria, analyzing dimensions including argument coherence, conceptual accuracy, citation quality, structural organization, and linguistic precision, thereby streamlining grading workflows while providing rapid feedback to learners at unprecedented scale.
- **Intelligent Content Recommendation Engine:** Machine learning models analyze comprehensive learner data including course completion patterns, assessment performance, time-on-task metrics, content interaction behaviors, stated career objectives, and demographic characteristics to generate personalized course recommendations that align with individual learning trajectories and professional development goals.
- **Adaptive Learning Path Optimization:** AI-driven systems dynamically adjust content sequencing, difficulty progression, and supplementary resource allocation based on real-time learner performance analytics, identifying knowledge gaps and automatically recommending targeted remedial content or advanced materials to optimize individual learning efficiency.
- **Predictive Analytics for Learner Success:** Algorithmic models process behavioral engagement data to identify learners at elevated risk of course abandonment, triggering proactive intervention strategies including personalized encouragement communications, peer study group recommendations, or supplementary support resource suggestions to enhance completion probabilities.

3.2 Udemmy

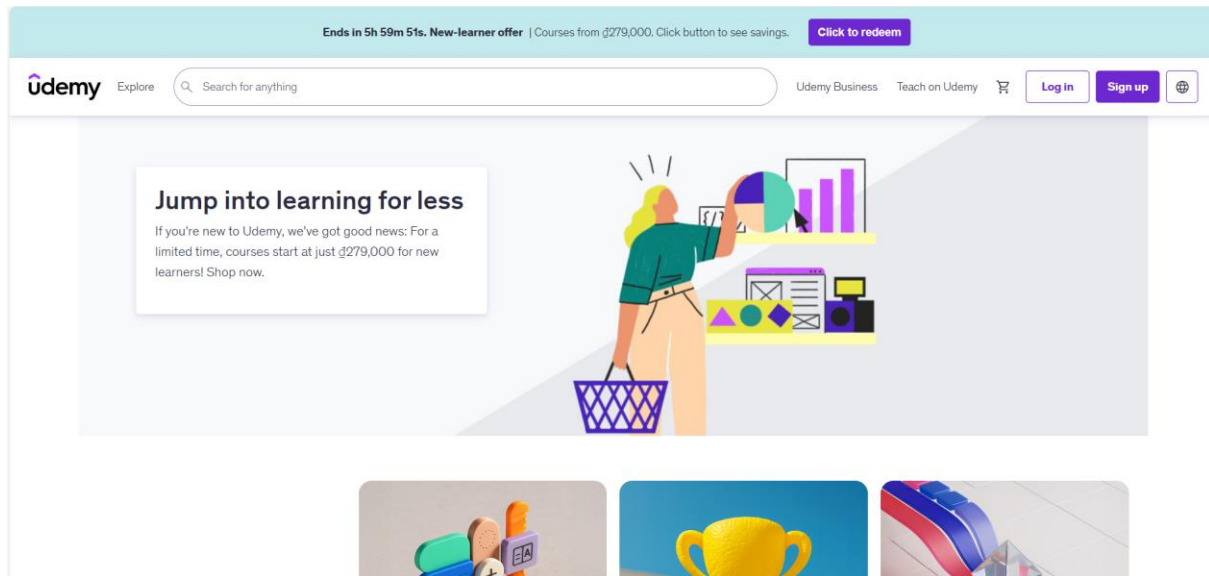


Figure 3.2: Udemmy

3.2.1 Description

Udemmy, founded in 2010 by Turkish entrepreneur Eren Bali alongside co-founders Oktay Caglar and Gagan Biyani, operates as a democratized global online education marketplace that fundamentally disrupts traditional educational gatekeeping mechanisms. The platform functions on a peer-to-peer instructional model, enabling individual practitioners, industry professionals, academic experts, and passionate enthusiasts to independently design, produce, and monetize educational courses across virtually unlimited subject domains. Learners access this extensive repository—exceeding 220,000 courses spanning categories from advanced software development and data science to creative arts, personal wellness, and niche hobby instruction—at their convenience through flexible, self-paced learning modalities. This decentralized content generation paradigm effectively bridges global knowledge holders with learners worldwide, transcending geographical, institutional, and socioeconomic barriers that characterize conventional educational systems.

Link: <https://www.udemy.com>

3.2.2 System Actors

The Udemmy ecosystem comprises diverse stakeholder groups whose interactions constitute the platform's operational dynamics:

- **Learners/Students:** End-users who browse the extensive course catalog using sophisticated search and filtering mechanisms, enroll in courses aligned with personal or professional development objectives, engage with multimedia instructional content at individualized pacing, complete assessments and practical exercises, post questions and participate in course-specific discussion forums, provide ratings and reviews influencing course discoverability, and ultimately apply acquired competencies in real-world contexts.
- **Instructors/Teachers/Content Creators:** Subject matter experts, industry practitioners, academic professionals, or skilled hobbyists who conceptualize course curricula, produce video lectures using personal recording equipment, design supplementary learning materials

including worksheets and reference documents, create assessment instruments, respond to learner inquiries within course discussion forums, monitor performance analytics dashboards tracking enrollment metrics and revenue generation, iterate course content based on learner feedback, and build personal brands through teaching reputation.

- **Platform Administrators:** Technical operations teams responsible for maintaining robust infrastructure capable of hosting massive video content libraries, implementing sophisticated content delivery networks ensuring global accessibility, enforcing community guidelines and content policies, moderating disputes between instructors and learners, providing technical support for both instructor content creation and learner consumption challenges, processing financial transactions including payment disbursement to instructors, and continuously optimizing platform functionality through user experience research and iterative design improvements.
- **Corporate Clients/Enterprise Users:** Organizations leveraging Udemy Business—the platform's enterprise-focused offering—for systematic employee upskilling initiatives, onboarding programs, compliance training, leadership development, technical certification preparation, and continuous learning culture cultivation, typically through volume licensing agreements providing curated course collections aligned with organizational competency frameworks.

3.2.3 Features

Udemy's feature ecosystem reflects its marketplace philosophy prioritizing diversity, accessibility, and instructor autonomy:

- **Expansive Course Marketplace:** Unparalleled repository exceeding 220,000 courses spanning 13 primary categories and hundreds of subcategories, encompassing development (programming languages, web development, mobile development, game development), business (entrepreneurship, finance, sales), IT and software (certification preparation, network security, operating systems), design (graphic design, UX/UI, 3D modeling), marketing (digital marketing, content marketing, SEO), personal development (productivity, leadership, career development), photography, music, health and fitness, teaching and academics, lifestyle, and numerous niche specializations rarely available through traditional educational channels.
- **Multimedia Instructional Content:** Predominantly video-driven pedagogical approach featuring instructor-led lectures ranging from brief conceptual explanations to comprehensive multi-hour workshops, enhanced by supplementary materials including quizzes testing comprehension, coding exercises with solution walkthroughs, downloadable resources such as PDFs, templates, code repositories, and reference sheets, and in some advanced courses, interactive coding environments or simulation tools embedded directly within the learning interface.
- **Perpetual Access Model:** Upon course purchase, learners receive lifetime access rights enabling unlimited content review, allowing individuals to reference materials months or years post-completion when applying concepts in evolving professional contexts, downloading videos for offline viewing during travel or in bandwidth-constrained environments, and revisiting challenging concepts at their own pace without temporal constraints imposed by subscription expiration.
- **Cross-Platform Accessibility:** Seamless learning experiences across web browsers supporting responsive design adapting to various screen dimensions, native mobile applications for iOS and Android featuring offline download capabilities allowing course consumption without

internet connectivity, progress synchronization ensuring continuity across devices, and ChromeCast/AirPlay integration enabling television-based viewing for enhanced visual comfort during extended learning sessions.

- **Digital Credential System:** Automated certificate generation upon course completion showcasing learner name, course title, completion date, and unique verification code, with LinkedIn integration facilitating professional network credential sharing, though certificates lack formal accreditation or academic credit value recognized by traditional educational institutions.
- **Instructor Toolkit Suite:** Comprehensive course creation infrastructure including video upload and processing capabilities supporting various file formats and resolutions, drag-and-drop curriculum builders organizing content into logical sections and lectures, quiz and assessment creation interfaces with multiple question formats, promotional coupon generation systems enabling instructors to implement marketing campaigns, performance analytics dashboards visualizing enrollment trends, revenue metrics, student engagement patterns, and review sentiment analysis.
- **Corporate Learning Solution (Udemy Business):** Enterprise subscription offering providing organizations with curated collections of high-quality courses vetted for business relevance, learning path recommendations aligned with common job roles and competency frameworks, administrative dashboards tracking employee engagement and skill development progress, integration capabilities with learning management systems (LMS) and human resource information systems (HRIS), and custom content upload functionality enabling proprietary training materials alongside marketplace courses.
- **Multilingual Global Reach:** Courses available in over 75 languages reflecting truly international instructor and learner communities, with platform interface localization supporting dozens of languages, subtitle and closed caption options enhancing accessibility for non-native speakers and hearing-impaired learners, and culturally diverse content perspectives challenging Western-centric educational paradigms.
- **Dynamic Pricing and Promotional Ecosystem:** Flexible instructor-controlled pricing strategies, frequent platform-wide promotional events offering courses at substantial discounts (often 80-95% reductions from list prices), bundle deals encouraging multi-course purchases, subscription models (Udemy Personal Plan) providing unlimited access to curated course subsets, and organizational volume licensing with negotiated enterprise pricing structures.

3.2.4 Pros

Udemy's marketplace model generates distinctive competitive advantages across accessibility, diversity, and practical orientation dimensions:

- **Unprecedented Course Diversity and Niche Specialization:** The open marketplace philosophy democratizing content creation has generated an extraordinarily comprehensive course catalogue addressing highly specialized topics, emerging technologies, regional skill requirements, and unconventional subject combinations rarely economically viable for traditional institutions, enabling learners to pursue highly individualized educational paths aligned with unique career trajectories or personal interests beyond mainstream offerings.
- **Exceptional Cost Accessibility:** Aggressive promotional pricing strategies rendering most courses available at \$10-\$30 during frequent sales events, one-time purchase models eliminating ongoing subscription commitments, free course options providing entry-level skill exploration, and organizational bulk purchasing discounts significantly reduce financial

barriers to quality education compared to premium platforms like Coursera or traditional institutions, democratizing learning opportunities across socioeconomic strata globally.

- **Practical, Application-Oriented Pedagogy:** Instructor backgrounds predominantly rooted in professional practice rather than academic theory cultivate courses emphasizing immediately implementable skills, real-world case studies, industry-standard tool proficiencies, and pragmatic problem-solving approaches that translate directly to workplace performance improvements, particularly valuable for learners seeking rapid skill acquisition for career transitions or job function enhancements rather than comprehensive theoretical foundations.
- **Flexible Self-Directed Learning:** Absence of enrollment deadlines, cohort-based pacing requirements, or mandatory interaction schedules accommodates learners with irregular availability, demanding professional commitments, caregiver responsibilities, or personal preferences for solitary study, empowering individuals to construct personalized learning experiences aligned with their lifestyles rather than conforming to institutionally imposed temporal structures.
- **Instructor Diversity and Authenticity:** Direct connections between working professionals and learners foster authentic knowledge transfer reflecting current industry practices, emerging trend awareness, practical troubleshooting wisdom accumulated through professional experience, and insider perspectives on career development pathways that academic instructors may lack due to institutional distance from contemporary workplace realities.
- **Rapid Content Currency:** The decentralized content creation model enables exceptionally rapid course development responding to emerging technologies, trending skills, breaking industry developments, or newly released software tools, often manifesting courses within weeks of significant market events compared to traditional educational institutions' multi-year curriculum development cycles constrained by bureaucratic approval processes.

3.2.5 Cons

Despite marketplace strengths, Udemy's openness generates significant quality assurance, credentialing, and engagement challenges:

- **Substantial Quality Inconsistency:** The minimal barriers to instructor participation and absence of rigorous content vetting create dramatic variance in pedagogical effectiveness, instructional design sophistication, content accuracy, production quality, and learning outcome achievement across the platform, requiring learners to invest considerable effort evaluating course quality through review analysis, preview content examination, and instructor reputation research before enrollment decisions, with significant risk of suboptimal learning experiences.
- **Limited Credential Market Recognition:** Certificates of completion lack institutional backing, formal accreditation, academic credit transferability, or professional licensure recognition, substantially diminishing their value in credential-sensitive employment contexts, graduate school applications, or professional certification pathways compared to university-affiliated or industry-standard certifications, positioning Udemy primarily as a skill development platform rather than formal credential acquisition mechanism.
- **Minimal Instructor-Learner Interaction:** The asynchronous, self-paced learning model combined with instructors' variable engagement levels typically results in limited opportunities for personalized feedback, complex question clarification, mentorship relationships, or adaptive instruction responding to individual learning difficulties, potentially

disadvantaging learners who require interactive pedagogical support or struggle with self-directed learning approaches requiring high intrinsic motivation.

- **Course Discovery and Selection Challenges:** The overwhelming volume of available courses creates decision paralysis, with hundreds or thousands of options addressing similar topics but varying dramatically in quality, approach, currency, and comprehensiveness, requiring substantial time investment in comparative evaluation while risking suboptimal selections that inefficiently allocate learners' limited time and financial resources.
- **Instructor Support Variability:** Post-purchase learner assistance depends entirely on individual instructor responsiveness, availability, expertise in addressing learner challenges, and ongoing course maintenance commitment, creating inconsistent experiences ranging from highly engaged instructors providing detailed personalized assistance to essentially abandoned courses lacking any meaningful post-publication support or content updates addressing learner feedback.
- **Content Currency Maintenance Issues:** Many courses become outdated as technologies evolve, software versions advance, industry practices shift, or regulatory frameworks change, with instructors lacking systematic incentives or obligations to maintain content currency through regular updates, potentially leaving learners with obsolete knowledge or deprecated skill sets despite recent enrollment.
- **Algorithmic Visibility Inequities:** Course discoverability heavily influenced by platform recommendation algorithms prioritizing metrics like enrollment volume, review ratings, revenue generation, and instructor promotional spending creates competitive advantages for established instructors while hindering discovery of newer, potentially higher-quality courses from emerging educators, potentially perpetuating mediocrity through network effects favoring popular rather than optimal content.

3.2.6 AI Implementation:

Udemy has demonstrated notably limited integration of advanced artificial intelligence technologies compared to competitors, with AI implementation remaining rudimentary relative to platform scale:

- **Basic Recommendation Systems:** Machine learning algorithms analyze learner browsing patterns, purchase history, course completion data, and demographic characteristics to generate personalized course suggestions appearing on homepage interfaces and within category pages, though recommendation sophistication appears limited compared to platforms like Coursera, often relying on collaborative filtering approaches emphasizing popularity metrics over nuanced skill gap analysis or career trajectory alignment.
- **Search Relevance Optimization:** Natural language processing techniques enhance search functionality by parsing course descriptions, instructor qualifications, learner reviews, and curriculum content to match user queries with relevant courses, implementing semantic understanding beyond simple keyword matching to surface contextually appropriate results even when exact terminology differs between searches and course metadata.

Notable Absence of Advanced AI Features: Unlike competitors implementing AI-powered assessment grading, adaptive learning path customization, predictive analytics for learner success, intelligent tutoring systems, or automated content quality evaluation, Udemy has not publicly deployed comparable sophisticated AI capabilities, potentially reflecting strategic prioritization of marketplace expansion and content volume over pedagogical innovation through artificial intelligence, or

technological infrastructure limitations constraining advanced AI implementation across the massive, heterogeneous course catalog.

3.3. NativeX



Figure 3.3: NativeX

3.3.1 Description

NativeX represents a strategically focused, geographically specialized online English language learning platform operating exclusively within the Vietnamese educational market. The platform concentrates on addressing the specific linguistic competency requirements of Vietnamese working professionals navigating increasingly globalized business environments where English proficiency constitutes a critical career advancement determinant. Unlike generalist language platforms offering broad demographic appeal across multiple languages and proficiency levels, NativeX has cultivated deep expertise in workplace English communication for Vietnamese learners, incorporating sophisticated understanding of Vietnamese-English linguistic interference patterns, cultural communication norms, professional context requirements, and Vietnamese learners' specific pedagogical preferences shaped by educational system characteristics.

Link: <https://nativex.edu.vn/>

3.3.2 System Actors

The NativeX ecosystem comprises specialized stakeholder groups reflecting its focused pedagogical mission:

- **Learners/Students:** Predominantly Vietnamese working professionals across diverse industries and career stages—including corporate employees in multinational companies, entrepreneurs engaged in international business development, technical professionals collaborating with global teams, aspiring managers seeking promotional advancement requiring English proficiency, job seekers targeting foreign companies or international assignments, and established professionals pursuing career transitions into English-intensive

roles—all united by practical workplace English acquisition objectives rather than academic language study motivations.

- **Teachers/Instructors:** Rigorously vetted international educators holding recognized professional teaching qualifications including TEFL (Teaching English as a Foreign Language), TESOL (Teaching English to Speakers of Other Languages), CELTA (Certificate in English Language Teaching to Adults), or equivalent certifications demonstrating pedagogical expertise in adult language instruction, combined with substantial experience teaching Vietnamese learners specifically, understanding cultural communication patterns, common linguistic challenges arising from Vietnamese-English structural differences, and workplace English instructional methodologies emphasizing practical communication competencies over purely academic language knowledge.
- **Content Providers and Curriculum Partners:** Established international educational publishers with proven expertise in English language teaching materials, including Macmillan Education—a leading global publisher with over 175 years of educational content development experience—and National Geographic Learning, renowned for authentic, engaging content incorporating real-world imagery, compelling narratives, and culturally diverse perspectives, ensuring curriculum quality, pedagogical soundness, alignment with international proficiency frameworks (CEFR), and professional relevance through workplace-contextualized learning materials.
- **Technology and Platform Development Teams:** Software engineers, user experience designers, educational technology specialists, and AI researchers responsible for building and maintaining the digital learning infrastructure, developing adaptive learning algorithms, creating interactive practice tools, implementing video conferencing capabilities supporting live instruction, ensuring platform reliability and security, and continuously innovating features enhancing learning effectiveness and engagement.

3.3.3 Features

NativeX's feature architecture reflects sophisticated integration of personalized learning design, live human instruction, scheduling flexibility, and AI-augmented practice:

- **Comprehensive Personalized Learning Pathways:** Learning journeys initiated through complimentary CEFR-aligned (Common European Framework of Reference for Languages) proficiency assessment examining reading comprehension, listening comprehension, grammatical accuracy, vocabulary range, and communicative competence across six internationally recognized levels (A1-C2), followed by detailed diagnostic analysis identifying specific linguistic strengths and developmental areas, culminating in customized learning roadmaps with clearly defined milestones, skill-specific objectives, and projected timelines aligned with learners' stated goals—whether professional presentation excellence, negotiation fluency, email communication proficiency, meeting participation confidence, or comprehensive business English mastery.
- **Live Interactive Small-Group Instruction:** Synchronous learning sessions featuring qualified international teachers conducting intimate classes limited to 1-6 participants maximum, dramatically exceeding typical language learning platforms' larger cohorts, thereby enabling substantial individual speaking practice time, personalized error correction, customized feedback addressing each learner's specific challenges, peer interaction facilitating conversational confidence development, and authentic communicative exchanges

emphasizing real-world workplace scenarios including presentations, negotiations, problem-solving discussions, team collaboration, and professional networking conversations rather than artificial textbook dialogues.

- **Exceptional Scheduling Flexibility:** Comprehensive class availability spanning 8:00 AM to 11:00 PM daily including weekends, accommodating Vietnamese professionals' demanding work schedules, commuting constraints, family responsibilities, and personal time preferences, with user-controlled scheduling interfaces enabling learners to book sessions aligned with their availability, modify bookings with reasonable advance notice, and maintain learning consistency despite irregular professional commitments through diverse time slot options eliminating common barriers to sustained language study participation.
- **AI-Powered Independent Practice Ecosystem:** Sophisticated artificial intelligence tools supplementing live instruction through self-directed practice activities including adaptive vocabulary exercises dynamically adjusting difficulty based on learner performance, grammar drills targeting common Vietnamese learner errors, pronunciation training utilizing speech recognition technology analysing phonetic accuracy, listening comprehension exercises featuring authentic workplace audio scenarios, reading comprehension passages aligned with professional contexts, and writing practice with automated feedback—all algorithmically personalized to address individual learning gaps identified through continuous performance monitoring and adapting to accelerate proficiency development.
- **Progress Tracking and Analytics Dashboard:** Comprehensive learner interfaces visualizing skill development trajectories across multiple linguistic dimensions, displaying achievement metrics relative to personalized goals, highlighting mastery levels for specific competency areas, identifying persistent challenge domains requiring focused attention, and providing motivational reinforcement through tangible progress documentation encouraging sustained engagement throughout potentially lengthy language acquisition journeys.
- **Professional Context Integration:** Curriculum design explicitly incorporating authentic workplace communication scenarios relevant to Vietnamese business contexts, including email correspondence conventions in international business, presentation structures and delivery techniques for multinational audiences, negotiation language and cultural considerations, meeting participation protocols, intercultural communication strategies, industry-specific vocabulary across common professional domains, and pragmatic language features (politeness strategies, indirect communication, appropriate formality levels) essential for effective professional relationship building.

3.3.4 Pros

NativeX's specialized positioning generates distinctive competitive advantages particularly salient for its target demographic:

- **Deep Cultural-Linguistic Contextualization:** Profound understanding of Vietnamese learners' first language characteristics enables precise identification and systematic remediation of predictable interference patterns (tonal language speakers learning non-tonal English, Vietnamese grammatical structures influencing English production, pronunciation challenges specific to Vietnamese phonological inventory), cultural communication norm differences requiring explicit instruction (directness levels, hierarchical relationship navigation, face-saving considerations), and learning style preferences shaped by Vietnamese educational traditions (explicit grammar instruction appreciation, structured curricula preference, teacher

authority respect), resulting in dramatically accelerated learning efficiency compared to generalist platforms lacking this specialized pedagogical knowledge.

- **Unparalleled Scheduling Flexibility and Accessibility:** Extended daily availability window (15 hours) combined with seven-day-per-week access accommodates the demanding professional commitments characteristic of Vietnamese corporate environments where long working hours, irregular schedules, and weekend work frequently constrain educational participation, eliminating temporal barriers that render conventional fixed-schedule language programs incompatible with learners' professional lives, thereby sustaining engagement and learning consistency critical for language acquisition success.
- **Enhanced Engagement Through Small-Group Interaction:** Intimate class sizes dramatically increase individual speaking practice time—often 15-20 minutes per one-hour session compared to 2-3 minutes in typical 15-20 person classes—accelerating conversational fluency development through substantially greater active production opportunities, immediate personalized correction, customized feedback addressing each learner's persistent errors, reduced performance anxiety in supportive small-group environments, and authentic peer interaction building communicative confidence essential for workplace English application.
- **Assured Quality Through Rigorous Teacher Credentialing:** Mandatory international teaching certifications (TEFL/TESOL/CELTA) combined with specific Vietnamese learner experience requirements ensure consistently high instructional quality, pedagogical expertise in adult language teaching, awareness of common Vietnamese learner challenges, cultural sensitivity, and professional conduct standards, substantially reducing quality variance plaguing platforms with minimal instructor vetting while building learner trust through transparent credentialing transparency.
- **Professionally Relevant Curriculum Alignment:** Exclusive focus on workplace English communication directly addresses Vietnamese professionals' primary motivation—career advancement through enhanced English proficiency—by concentrating instructional time on immediately applicable skills (business presentations, email writing, meeting participation, negotiation language) rather than diluting focus across general conversational topics, academic writing, or examination preparation irrelevant to working professionals' pragmatic objectives, thereby maximizing return on learners' limited time investment.
- **Comprehensive Learning Ecosystem Integration:** Synergistic combination of live instructor-led sessions providing human interaction, personalized feedback, and conversational practice with AI-powered independent study tools enabling flexible skill reinforcement, targeted weakness remediation, and continuous practice beyond scheduled classes creates a holistic learning architecture addressing language acquisition's multifaceted requirements more comprehensively than platforms offering exclusively live or exclusively self-study modalities.

3.3.5 Cons

Despite specialization advantages, NativeX confronts strategic limitations constraining scalability and competitive resilience:

- **Narrow Demographic Targeting and Market Constraints:** Exclusive focus on Vietnamese working professionals seeking workplace English proficiency fundamentally limits total addressable market size to a specific geographical region and demographic segment, constraining revenue potential, restricting international expansion opportunities, limiting cross-market learning and operational efficiencies, and creating vulnerability to Vietnamese

economic fluctuations, regulatory changes affecting education sector, or shifts in English proficiency priorities that could dramatically impact demand concentration in a single market.

- **Premium Pricing Positioning Limiting Accessibility:** Higher pricing relative to alternative language learning platforms (generalist apps offering \$10-20 monthly subscriptions, YouTube-based free content, community-based language exchange) creates accessibility barriers for price-sensitive segments including students, early-career professionals, small business owners, or individuals in smaller Vietnamese cities with lower income levels, potentially constraining market penetration and limiting the platform's impact on broader Vietnamese English proficiency improvement despite superior pedagogical effectiveness for those who can afford premium pricing.

3.3.6 AI Implementation

- **Advanced AI-Driven Pronunciation Analysis and Feedback Systems:** Sophisticated speech recognition algorithms utilizing deep learning models trained on diverse English accents and Vietnamese-accented English specifically analyses learner audio recordings across multiple phonetic dimensions including individual phoneme accuracy (particularly problematic sounds for Vietnamese speakers like /θ/, /ð/, /v/, final consonants), word-level stress patterns (critical for intelligibility), sentence-level intonation contours (distinguishing statements, questions, emphasis), rhythm and timing characteristics (syllable-timed Vietnamese versus stress-timed English), and overall intelligibility scoring, generating comprehensive feedback including numerical pronunciation accuracy scores, visual phonetic representations highlighting specific articulation errors, audio comparisons juxtaposing learner production with native speaker models, and actionable improvement recommendations with targeted practice exercises addressing identified deficiencies.

4. Business Opportunity

Vietnam's EdTech market is projected to reach significant valuation by 2029, fuelled by a digital economy demanding continuous upskilling. The shift from 'degree-based' to 'skill-based' hiring creates an urgent need for short-term, high-impact courses.

SkillUp captures this opportunity through a hybrid value proposition:

- For Learners: A frictionless path from discovery to mastery with localized support and payments.
- For Instructors: A 'Business-in-a-Box' solution. Instead of building their own websites, instructors use SkillUp's infrastructure (hosting, payment, LMS) to monetize their knowledge instantly.
- Scalability: The platform model allows for exponential content growth without proportional increases in operational costs, ensuring long-term sustainability.

The online learning market in Vietnam is rapidly growing, fuelled by widespread Internet adoption, the demand for flexible learning, and a strong cultural emphasis on continuous skill development. According to the [Vietnam Edtech Market Summary, Competitive Analysis and Forecast to 2029](#), the sector is poised for significant expansion, driven by government initiatives to digitize education and a rising middle class willing to invest in quality learning resources. Projections indicate that the number of users in the online learning platform market alone could reach approximately 11.8 million by 2029, highlighting a massive addressable market.

Despite this growth, a significant gap remains. Both students and working professionals are actively seeking a user-friendly platform that is not only affordable but also provides practical, applicable knowledge tailored to the local context. SkillUp is uniquely positioned to bridge this gap. By offering intuitive tools for instructors to easily create and monetize courses, and a localized interface for learners, SkillUp fosters a vibrant, interactive learning community. Beyond serving individual learners, the platform has the potential to capture the B2B segment by expanding into corporate training, helping organizations improve workforce quality at optimized costs in this booming digital economy.

5. Software Product Vision

SkillUp is an online learning platform designed to help Vietnamese learners easily access high-quality knowledge, develop practical skills, and expand career opportunities. The product delivers a personalized, flexible, and effective learning experience for individuals, while supporting instructors and organizations in managing and measuring training programs. The vision of SkillUp is to become the leading digital learning ecosystem in Vietnam, where advanced technology goes hand in hand with human development, contributing to a dynamic and sustainable learning community.

6. Project Scope & Limitations

6.1 Major Features

No.	ID	Feature	Description
1	FE-01	Common	Login: Users can log in using email/password or Google OAuth. The system validates credentials and generates JWT tokens for authenticated sessions..
			Register: New users can register by providing email, password, and personal information. Email verification via OTP is required to activate the account.
			Change Password: Authenticated users can change their password by providing their current password and a new password.
			Forgot Password: Users who forget their password can request a reset link via email. The system sends a secure token to reset the password.
			Notification: Users receive real-time notifications about course updates, payment confirmations, and system announcements via SignalR hub.
			Profile Management: Users view their personal information, can edit, update information or change passwords
2	FE-02	Student's Features	View Course: view course details including curriculum, lecturer info, ratings, and preview lessons before enrollment.
			Learning Course: enrolled courses with structured sections and lessons. Video playback with progress tracking , and lesson completion status. The system provides instant grading with multiple-choice and multiple-answer questions. Results show correct/incorrect answers and score percentage.
			Learning progress tracking: system automatically tracks student progress across sections , lessons , and quizzes.
			Forum & Community: Students can create forum posts, comment on posts, like/unlike comments, and engage in course-related discussions.
			AI chat in lesson : Students can chat with an AI assistant while watching lessons to ask questions, clarify doubts, and get instant responses related to course content.
			Shopping cart & payment: add multiple courses to cart, apply vouchers for discounts, and complete purchases via PayOS payment gateway.
3	FE-03	Lecturer's Feature	Apply as Lecturer : apply to become a lecturer by submitting CV (PDF) and degree certificates (images). Applications are reviewed by System Moderators.
			Course Management : Lecturers can create new courses with details, edit course information, and manage course status.
			Quiz Management : Lecturers create quizzes within sections, assign questions from question bank.
			Student Analytics : Lecturers view enrolled students, track their progress, quiz scores, completion rates, and student engagement metrics per course.
			Question Management : Lecturers create question pools with multiple-choice or multiple-answer questions, add answer options, mark correct answers, and attach images if needed. Questions are reusable across quizzes.

			Revenue and income report: Lecturers access revenue reports showing total earnings, course sales breakdown, and financial analytics with charts.
4	FE-04	System Moderator's Features	Lecturer Application Review: System Moderators review lecturer applications including CV and degree certificates. They can accept applications (converting Student to Lecturer role and setting account status to Active) or reject with reasons via email notification.
			User Account Management: Moderators view all user accounts across roles, search/filter users, view user details, manage account status (Active/Pending/Banned), and monitor user activities for security purposes.
			Ticket Management: System Moderators handle support tickets from lecturers and students. They can view ticket details, provide solutions, escalate issues, mark tickets as resolved, and track resolution time. AI suggests ticket titles for faster categorization.
			System Moderator Dashboard: Central dashboard displaying platform statistics: total users by role, pending lecturer applications, unresolved tickets, recent user activities, and system health metrics.
5	FE-05	Content Moderator's Features	Course Approval: Content Moderators review course submissions from lecturers. They can approve courses (making them public), reject with reasons, or request modifications. Quality checks include content accuracy and appropriateness.
			News Management: Moderators create news articles using TinyMCE editor with rich formatting, upload images, publish/unpublish news, edit existing articles, and manage news visibility on platform.
			Forum Post Moderation: Moderators monitor forum posts and comments for inappropriate content, spam, or violations. They can delete posts, ban users temporarily, and enforce community guidelines.
			Category Management: Moderators create, edit, or delete course categories and subcategories. Categories organize courses for easier browsing and help students find relevant content.
			Report Comment Management: Moderators review reported comments from students/lecturers. They investigate reports, take action (delete comment, warn user), and resolve report status.
			Content Moderator Dashboard: Central dashboard providing an overview of platform content and moderation status. It displays key content metrics, tracks course approval status (Public, Pending Approval, Hidden, Reported), and highlights pending tasks, such as unprocessed reports. It also shows recent publishing activity.
6	FE-06	Admin's Features	Banner Management: Moderators upload promotional banners for homepage, set banner priority/order, configure display duration, and manage banner visibility for marketing campaigns.
			Moderator Management: Admins create Content Moderator and System Moderator accounts, assign permissions, manage moderator roles, deactivate accounts, and send account credentials via email.
			Financial Reports: Admins generate lecturer salary reports based on course sales, calculate platform commission, export financial data, track payment transactions, and manage revenue distribution
			Platform Notifications: Admins broadcast system-wide notifications to all users or specific roles (e.g., maintenance announcements, policy updates, promotional messages). Notifications support rich text formatting.

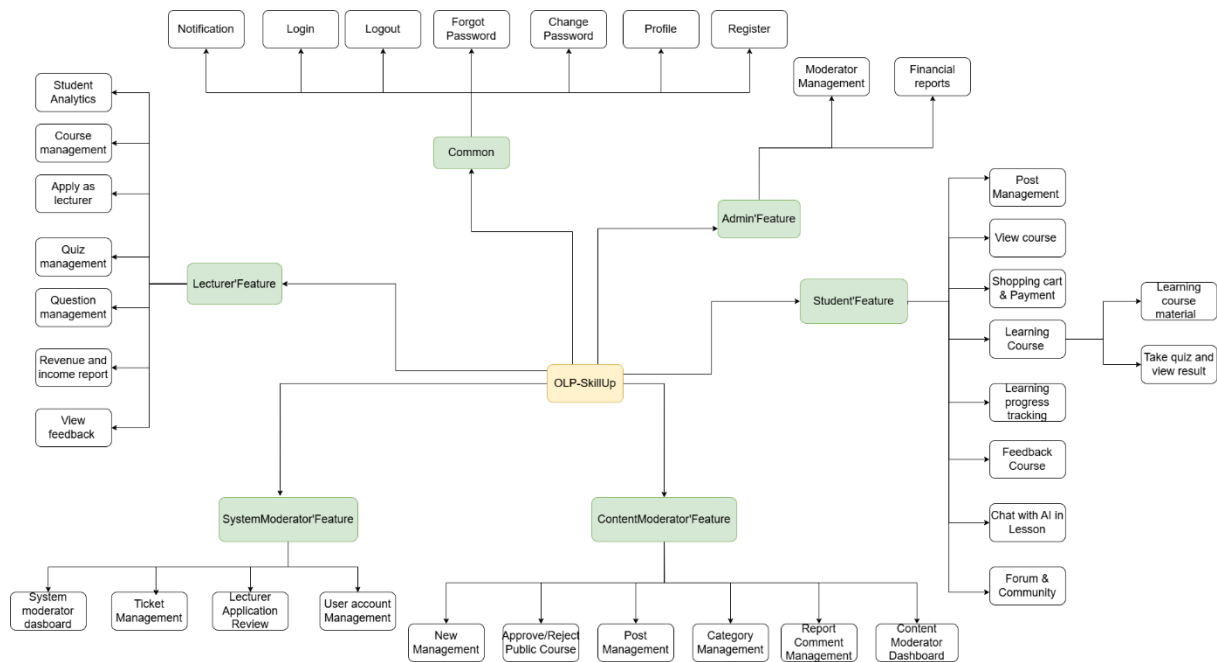


Figure 6.1: Major Features

6.2 Limitations & Exclusions

- LI-1: Communication with the instructor will not be optimal, so answering questions or giving instructions will take time to resolve.
- LI-2: A large number of learners asking questions to one lecturer will significantly reduce the learning efficiency.
- LI-3: Answers from the AI feature may not be accurate and cannot be guaranteed to be accurate.
- LI-4: The system is currently integrated with and uses the Gemini API through an API key provided by a third party. Therefore, the team does not have full control over service stability, usage limits, response speed, or potential policy changes from the API provider.